a

Energy

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Broad, overlapping components:
coarser readout

Deep, isolated wells:
exact readout

Number-word frequency

1.0 0.8 0.6 0.4 0.2 0.0

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Evidence axis x

b

Final state x^*

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

$y_0=2$

1 2 3

$y_0=6$

5 6 7 8 9

Static readout
reference

Readout distribution
over integers

$y_0=13$

11 13 15

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Initial estimate y_0

c

Predicted RT (s)

2.5 2.0 1.5

Basal latency t_0

Attractor time $\tau E[T]$

Fast, accurate

Slow, approximate

$\mathrm{RT} = t_0 + \tau E[T]$

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Presented numerosity n